Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Theory of Computation

CSIT — 4th Semester — 2081 — Answer Sheet

Subject Code: CSC262

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Mention the transition function of PDA. List the two ways that PDA accepts the string. Convert the following CFG to PDA.$S \rightarrow AS \mid \epsilon$$B \rightarrow aAb \mid Bb \mid ab$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. List any two regular operators. Minimize the following finite state machine using Table Filling algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define Turing machine as enumerators of strings of a language. Encode the Turing machine $TM: (\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, B\}, \delta, q_0, B, F)$, with input $w: ba$ and δ is defined as follows.$\delta(q_0, b) = (q_1, a, R)$, $\delta(q_1, a) = (q_1, b, R)$, $\delta(q_2, b) = (q_1, a, R)$, $\delta(q_1, B) = (q_2, b, L)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Does machine always refer to hardware? Justify. Define positive closure and kleene closure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is undecidable problem? Discuss about Post's Correspondence Problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define the language of a grammar. For the grammar $S \rightarrow 0S0 \mid 0 \mid 1 \mid \epsilon$, show the leftmost derivation for the string 00100 with its parse tree.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define \(\epsilon\) - closure of a state. Differentiate between Moore and Mealy machine.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Represent the following regular grammar to finite automata.\(S \rightarrow a \mid aA \mid bB \mid \epsilon\)\(A \rightarrow aA \mid aS\)\(B \rightarrow bS \mid \epsilon\)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Design the DFA that accepts binary string ending with "00" and show its extended transition function for the string 111000.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Convert the following grammar to CNF.\(S \rightarrow AAB\), \(A \rightarrow aAb \mid \epsilon\), \(B \rightarrow aB \mid a\)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. For the following Turing Machine, test whether the string "( )) (" is accepted or rejected and show it in transition()XYB\(q_0\)\((q_1, X, R)\)\((q_3, Y, R)\)\(q_1\)\((q_1, (, R)\)\((q_2, Y, L)\)\((q_0, X, R)\)\((q_1, Y, R)\)\(q_2\)\((q_2, (, L)\)\((q_0, X, R)\)\((q_2, Y, L)\)\(q_3\)\((q_3, Y, R)\)\((q_4, B, R)\)\(q_4\)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Differentiate between Class P and Class NP problem. Mention the transition function of DFA, NFA and \(\epsilon\) - NFA.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
